FINITE LOCALLY CONSTANT SHEAVES ON A LIMIT OF QUASI-COMPACT QUASI-SEPARATED SCHEMES FORM THE COLIMIT OF THE CATEGORIES ON THE APPROXIMANTS

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof is written out in full: the reduction of the three cases to essential surjectivity, the trivialising finite affine etale covering, its approximation on some X_{i'} and the spreading of the isomorphism of constructible sheaves to that stage.

Editorial note. The label lemma-category-loc-cst-is-colimit names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. The statement and proof are copied from the pinned chapter file; the two cited inputs are kept as external pointers with their canonical Stacks tags.

Editorial note. The preamble is the chapter's own native preamble excerpt already used for this source: the Xy-pic setup (native preamble lines 14-18) and the native theorem declarations with the mathematical macros (native preamble lines 154-217).

Allowed standard prerequisites (editorial).

- The limit presentation X = lim X_i with affine transition morphisms. Same chapter, the identification of the category of constructible sheaves on X as the colimit of those on the X_i, which the present lemma sharpens to locally constant sheaves. Supplies full faithfulness and the form of the remaining question, namely that a constructible sheaf becoming finite locally constant on X already does so at some stage.
- Approximation of an etale covering of the limit. Same chapter, the lemma allowing an etale covering of X with affine members to be approximated by an etale covering of some X_{i'}. Supplies the passage from the covering U of X trivialising F to a covering U_{i'} of X_{i'} with the same index set.

Source reference. Original reference 09YU: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 12552–12552. Prerequisite pointer invoked at proof line(s) 12652; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0EYL: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 7336–7336. Prerequisite pointer invoked at proof

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line(s) 12663; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let $X = \lim_{i \in I} X_i$ be a limit of a directed system of schemes with affine transition morphisms. We assume that X_i is quasi-compact and quasi-separated for all $i \in I$.*

(1) *The category of finite locally constant sheaves on $X_{étale}$ is the colimit of the categories of finite locally constant sheaves on $(X_i)_{étale}$.*
(2) *The category of finite locally constant abelian sheaves on $X_{étale}$ is the colimit of the categories of finite locally constant abelian sheaves on $(X_i)_{étale}$.*
(3) *Let Λ be a Noetherian ring. The category of finite type, locally constant sheaves of Λ-modules on $X_{étale}$ is the colimit of the categories of finite type, locally constant sheaves of Λ-modules on $(X_i)_{étale}$.*

Proof. By Lemma 09YU the functor in each case is fully faithful. By the same lemma, all we have to show to finish the proof in case (1) is the following: given a constructible sheaf $\mathcal{F}_i$ on X_i whose pullback $\mathcal{F}$ to X is finite locally constant, there exists an $i' \geq i$ such that the pullback $\mathcal{F}_{i'}$ of $\mathcal{F}_i$ to $X_{i'}$ is finite locally constant. By assumption there exists an étale covering $\mathcal{U} = \{U_j \to X\}_{j \in J}$ such that $\mathcal{F}|_{U_j} \cong \underline{S_j}$ for some finite set S_j. We may assume U_j is affine for all $j \in J$. Since X is quasi-compact, we may assume J finite. By Lemma 0EYL we can find an $i' \geq i$ and an étale covering $\mathcal{U}_{i'} = \{U_{i',j} \to X_{i'}\}_{j \in J}$ whose base change to X is $\mathcal{U}$. Then $\mathcal{F}_{i'}|_{U_{i',j}}$ and $\underline{S_j}$ are constructible sheaves on $(U_{i',j})_{étale}$ whose pullbacks to U_j are isomorphic. Hence after increasing i' we get that $\mathcal{F}_{i'}|_{U_{i',j}}$ and $\underline{S_j}$ are isomorphic. Thus $\mathcal{F}_{i'}$ is finite locally constant. The proof in cases (2) and (3) is exactly the same. □